

L13: Estimating the Number of Nodes with Traceroute-like Methods

Let us now see how we can use a traceroute-like sampling strategy to estimate the number of nodes in a large network. Suppose that we have a set of sources $\mathcal{S} = \{s_1, s_2, \dots, s_{n_s}\}$ and a set of targets $\mathcal{T} = \{t_1, t_2, \dots, t_{n_T}\}$. Each traceroute starts from a source node and it traverses a network path to a target node, also "observing" (sampling) the intermediate nodes in the path. Note that an intermediate node may be observed in more than one traceroute path. The number of observed nodes is denoted by N^* , while the number of nodes in the population network is denoted by N . How can we use N^* to estimate N ?

Here is the key idea in the following method: suppose that we drop a given target node t_j from the study. Would that target be observed in the traceroute paths to the remaining targets? We can easily measure the fraction of targets that would be observed in this manner. *Intuitively, the lower this fraction is, the larger the population network we are trying to sample.* Is there a way we can use this fraction to "inflate" N^* so that it gives us a good estimate of N ? As we will see next, the answer is yes through a simple mathematical argument.

Let us introduce some notation first. Suppose that $V_{(-j)}^*$ is the set of observed nodes when we drop t_j from the set of targets. The number of such nodes is $N_{(-j)}^* = |V_{(-j)}^*|$.

The binary variable δ_j is equal to one if t_j is NOT observed on sampled paths to any other target -- and zero otherwise. The total number of such targets (that can be observed only if we traceroute directly to them) is $X = \sum_j \delta_j$.

The probability that t_j is not observed on the paths to any other target, however, is simply the ratio between the number of nodes that are not observed after we remove t_j , and the number of non-source and non-target nodes:

$$P(\delta_j = 1 | V_{(-j)}^*) = \frac{N - N_{(-j)}^*}{N - n_s - (n_T - 1)}$$

assuming that the targets are chosen through random sampling without replacement from the set of all non-source nodes in the population graph.

The expected value of $N_{(-j)}^*$ is the same for all j however, simply due to symmetry (j can be any of the targets). Let us denote that expected value as $E[N_{(-j)}^*] = E[N_{(-)}^*]$.

So, the expected value of X is:

$$E[X] = \sum_{j=1}^{n_T} P(\delta_j = 1 | V_{(-j)}^*) = n_T \frac{N - E[N_{(-)}^*]}{N - n_s - (n_T - 1)}$$

We have now reached our goal: we can solve the last equation for the size of the population graph N :

$$N = \frac{n_T E[N_{(-)}^*] - (n_s + n_T - 1) E[X]}{n_T - E[X]}$$

The expected value $E[X]$ of the targets that can be observed only if we traceroute to them can be estimated by X , which is directly available from our traceroute data.

And the expected value $E[N_{(-)}^*]$ can be estimated, again directly from our traceroute data, as the average of $N_{(-j)}^*$ across all targets j .

Let us now see what happens when this method is applied to estimate the size of an older snapshot of the Internet that included $N=624,324$ nodes and 1,191,525 edges. The number of sources was $n_s = 10$. The "target density", defined as $\rho_T = n_T / N$, is shown as the x-axis variable in the following graph. The y-axis shows the fraction of the estimated network

size over the actual network size (ideally it should be equal to 1). The solid dots show the estimates from the method we described above (including intervals of $\pm$ one standard deviation around the mean). The open dots show the same fraction if we had simply estimated the size of the network based on N^* .

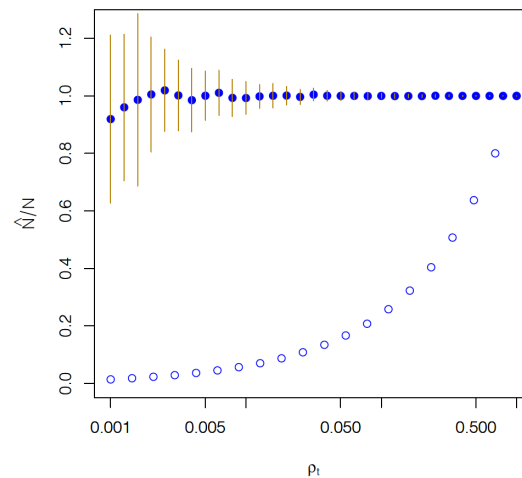


Image source: Kolaczyk, Eric D. *Statistical Analysis of Network Data: Methods and Models* (2009) Springer Science+Business Media LLC.

Note that the previous method is fairly accurate even if the target density is as low as 0.005 (i.e., about 3,100 targets).

On the contrary, if we had estimated the size of the network simply based on the number of observed nodes we would grossly underestimate how large the network is (unless if our targets pretty much cover all network nodes).

Food For Thought

If you could place the n_g sources anywhere you want, where would you place them to improve this estimation process?